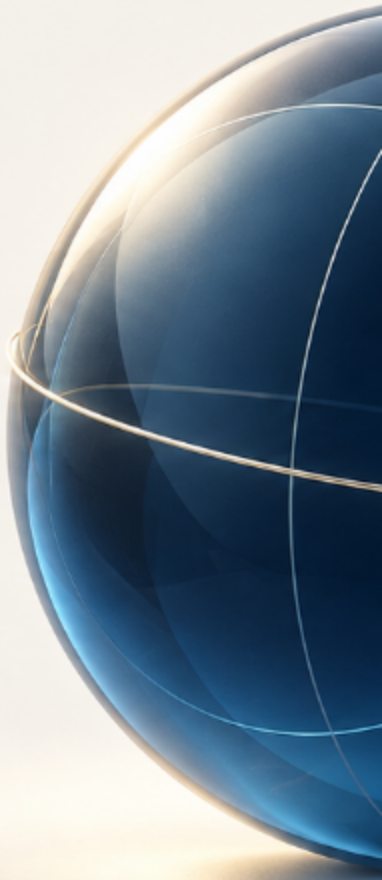
QUBIT ZERO TO HERO

Your first qubit simulation



S00 · Course orientation
Windows and Ubuntu setup

Course map and S00 outcomes

Foundations	Complex numbers, vectors, qubit states and gates
Dynamics	Hamiltonians, time evolution, noise and validation
Research bridge	Pulse control and a small QAOA assignment example
Today	Create an environment, run basics, explain its outputs

The local toolchain

Git + repository	Download and track the course source
Python + .venv	Run code in an isolated project environment
QuTiP 5.3.1	Represent quantum states and solve dynamics
Terminal + editor	Run commands and edit exercise files

Install Git and Python first. The package requires Python 3.11 or newer.

Windows setup · PowerShell

```
git clone https://github.com/croznnt/qubit-zero-to-hero-english.git
cd qubit-zero-to-hero-english
py -3.13 -m venv .venv
.\.venv\Scripts\python.exe -m pip install --upgrade pip
.\.venv\Scripts\python.exe -m pip install -e ".[dev]"
```

Run from the repository root, where pyproject.toml is located.

Ubuntu setup · Bash

```
sudo apt update
sudo apt install git python3 python3-venv python3-pip
git clone https://github.com/croznnt/qubit-zero-to-hero-english.git
cd qubit-zero-to-hero-english
python3 -m venv .venv
./.venv/bin/python -m pip install -e ".[dev]"
```

Use a native Ubuntu terminal. Keep pip installations inside .venv.

Interpreter and dependency check

Windows · PowerShell

```
.\.venv\Scripts\python.exe -c "import qutip; print(qutip.__version__)"
```

Ubuntu · Bash

```
./.venv/bin/python -c "import qutip; print(qutip.__version__)"
```

Expected QuTiP version: 5.3.1. Activation is optional with these paths.

Repository map

lessons/	Concept explanations and worked examples
exercises/	Tasks, starters and reference solutions
src/qubit_course/	Reusable models, experiments and command line tools
tests/	Automated checks of numerical behavior
runs/	Generated experiment outputs

Edit source files in the checkout. Treat .venv as disposable.

The four experiment artifacts

metrics.json	Summary values used to judge the result
metadata.json	Versions, source commit, platform and units
arrays.npz	Compressed NumPy arrays for further analysis
figure.png	A saved visual summary to open locally

Example run directory: runs/first/basics/
The random seed is recorded in metrics.json.

A reproducible run record

Source	Record the commit and whether the working tree is dirty
Environment	Keep Python and package versions with the result
Randomness	Record the seed and the number of simulated shots
Output	Choose a fresh output directory for each completed run

A fixed seed alone does not guarantee identical results across environments.

First experiment · basics

Windows · PowerShell

```
.\.venv\Scripts\python.exe -m qubit_course.cli basics `
  --seed 42 --output runs/first
```

Ubuntu · Bash

```
./.venv/bin/python -m qubit_course.cli basics \
  --seed 42 --output runs/first
```

Choose one operating system. Output: runs/first/basics/

Probability and sampled frequency

probability_one ≈ 0.25	The state's probability for outcome 1
Measured fraction = 0.262	Observed fraction in the seed 42 reference run
shots = 2000	Number of simulated binary measurements
bloch_norm ≈ 1	A consistency check for this pure state

Finite samples fluctuate. A different frequency does not by itself mean a bug.

Readiness checkpoint

Question 1	How does a repository differ from a virtual environment?
Question 2	Why can 0.262 be consistent with probability 0.25?
Question 3	Does an untouched starter reporting UNFINISHED mean setup failed?
Task 1	Record your checkout, interpreter and four artifact paths.
Task 2	Run seed 7 in runs/second and compare the metrics.

Next: S01 introduces complex numbers and the notation behind qubit states.

Appendix · Windows checks

```
.\.venv\Scripts\python.exe -m qubit_course.labs check basics
.\.venv\Scripts\python.exe -m qubit_course.labs check basics --solution
.\.venv\Scripts\python.exe -m pytest -q
```

UNFINISHED	Expected for the untouched learner starter
No module named qubit_course	Reinstall editable with this same interpreter
Activate.ps1 blocked	Continue with the explicit python.exe path

Appendix · Ubuntu checks

```
./.venv/bin/python -m qubit_course.labs check basics
./.venv/bin/python -m qubit_course.labs check basics --solution
./.venv/bin/python -m pytest -q
```

venv creation fails	Install python3-venv for the selected Python
Externally managed environment	Use .venv/bin/python for pip installation
No graphical display	Open the saved figure.png in an image viewer